MiNA – Financial Wealth (MiNA-Finw)

Technical Appendix and Data Description

Micro-based National Accounts for the Swedish Household Sector

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Abstract

This appendix documents the construction of a individual-year panel of financial wealth for the Swedish population, covering 1999–2015. Financial wealth comprises three components: mutual fund holdings, investment savings accounts (ISK), and directly held equities. Because Swedish administrative records do not directly report individual financial positions for all asset types and periods, each component requires a distinct estimation strategy. Fund holdings before 2012 are inferred from dividend payments reported on tax forms; from 2012 onward, fund and ISK values are recovered by using flat-rate tax assessments introduced with the 2012 tax reform. Equity holdings are observed directly in the security ownership register maintained by Euroclear Sweden and priced using a hierarchy of market data sources. The construction also produces weighted average returns of stocks and funds for each individual. The final dataset delivers individual-level fund wealth, ISK capital base, stock wealth, and the associated fund and stock portfolio returns.

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1 Introduction and Scope

Financial wealth—the value of mutual funds, equities, and tax-advantaged investment accounts held by individuals—is a major component of the Swedish household balance sheet. Measuring it at the individual level from administrative data poses two challenges. First, individual financial positions are not always directly observed: fund holdings before 2012 must be inferred indirectly from tax records, and ISK account values must be backed out from flat-rate tax assessments. Second, pricing directly held equities requires reconciling multiple market data sources with varying coverage across securities and time.

This appendix describes the construction of a person-year panel of financial wealth for the entire Swedish population over 1999–2015. The panel decomposes financial wealth into three sub-components—mutual funds, ISK accounts, and directly held stocks—and further returns to fund and stock holdings. The resulting dataset enters the household balance sheet as $W_{i,t}^{\text{Fin}}$, the financial wealth component of total net wealth.

A key structural break governs the construction: Sweden's 2012 tax reform eliminated fund-level dividend distributions and introduced flat-rate taxation of ISK accounts, fundamentally changing the information available in administrative records. The estimation strategy therefore differs across the pre- and post-reform periods.

Section 2 describes the data sources and sample construction. Section 3 introduces the key definitions and notation. Section 4 details the estimation procedure for each component. Section 5 describes the final dataset.

2 Data Sources and Sample Construction

2.1 Overview of Data Sources

The construction relies on three categories of data: administrative records, securities registry data, and commercial pricing databases.

Administrative records. We use administrative sources and Statistics Sweden. The KU31 tax form reported to the Swedish Tax Agency (*Skatteverket*) records dividends paid to each individual per fund ISIN for income years 1999–2015. This data is retrieved via the Income Tax Registry (*Inkomst- och Taxeringsregistret (IoT)*; Statistics Sweden (2016)), which also provides, from 2012 onward, the flat-rate standard income (*schablonintäkt*) on fund holdings and ISK accounts.

Securities registry. The Euroclear Sweden securities registry records individual-level equity holdings by ISIN (Euroclear Sweden, 2015a). Euroclear also publishes security master files linking ISINs to issuers and transaction-level price data Euroclear Sweden (2015b).

Pricing databases. The Finbas equity database, maintained by the Swedish House of Finance, provides daily bid prices for Swedish-listed securities (FinBas, 2016). Morningstar provides fund-level net asset values (NAV) and dividend-per-share data (Morningstar, 2009). The exchange rates are retrieved from Sveriges Riksbank (2025).

Table 1: Source data used in the construction

Source	Provider	Years	Information contributed
KU31 data via Income Tax Registry	Statistics Sweden (2016)	1999–2015	Fund dividends paid to each individual per ISIN
Income Tax Registry	Statistics Sweden (2016)	2012–2016	Flat-rate standard income on funds and ISK accounts
Euroclear holdings	Euroclear Sweden (2015a)	1999–2015	Monthly individual-level equity holdings by ISIN
Euroclear security master	Euroclear Sweden (2015b)	2001–2015	ISIN-to-issuer crosswalk and security classification
Finbas	FinBas (2016)	1999–2015	Daily bid prices for Swedish-listed equities
Morningstar	Morningstar (2009)	1999–2015	Fund NAV and dividend-per-share data
Exchange rates	Sveriges Riksbank (2025)	1999–2024	Year-end SEK exchange rates

2.2 Sample Definition and Construction

The sample comprises all individuals appearing in at least one of the three financial asset datasets—KU31 (fund dividends), IoT (flat-rate standard income), or Euroclear (equity holdings)—in a given year. The resulting dataset covers 1999–2015 for funds and stocks, 2012–2015 for ISK accounts, and 2000–2015 for the portfolio returns.[1] Individuals present in one component but absent from another receive zero values for the missing component, ensuring that total financial wealth is always the sum of its parts.

3 Key Definitions and Base Variables

Financial wealth for individual i in year t is defined as the sum of three components:

$$W_{i,t}^{\text{Fin}} = W_{i,t}^{\text{Fund}} + W_{i,t}^{\text{ISK}} + W_{i,t}^{\text{Stock}}, \tag{1}$$

where $W_{i,t}^{\text{Fund}}$ is the total value of mutual fund holdings, $W_{i,t}^{\text{ISK}}$ is the ISK capital base, and $W_{i,t}^{\text{Stock}}$ is the total value of directly held equities. All values are in nominal SEK at year-end.

For funds and stocks, we further calculate the yearly weighted returns of the individual's portfolio. The individual's portfolio-weighted return is:

$$r_{i,t}^{\text{Fin,port}} = \frac{\sum_j V_{i,j,t-1}^{\text{Fin}} \cdot r_{j,t}^{\text{Fin}}}{\sum_j V_{i,j,t-1}^{\text{Fin}}}. \tag{2}$$

We use the following additional notation throughout. Let $d_{j,t}^{\text{Fin}}$ denote the dividend per share of fund j in year t, and let $d_{i,j,t}^{\text{Fin,total}}$ denote the total dividend received by individual

[1] The return calculations require lagged values, so 1999 serves only as a base year.

i from fund j in year t. Let $\text{NAV}_{j,t}^{\text{Fin}}$ denote the year-end net asset value per share, and $V_{i,j,t}^{\text{Fund}} = n_{i,j,t}^{\text{Fund}} \cdot \text{NAV}_{j,t}^{\text{Fin}}$, where $n_{i,j,t}^{\text{Fund}}$ denote the number of fund shares held by individual i in fund j at the end of year t, and $n_{i,j,t}^{\text{Stock}}$ the corresponding count for directly held equities. Let $P_{j,t}^{\text{Fin}}$ the year-end equity price and $\text{SI}_{i,t}^{\text{Fund}}$ denote the flat-rate standard income on fund holdings and $\text{SI}_{i,t}^{\text{ISK}}$ the flat-rate standard income on ISK accounts, both as reported in the IoT register. Let σ_t^{ISK} denote the year-specific ISK standard-income rate, and let $\phi^{\text{Fund}} = 0.004$ denote the flat-rate standard-income rate on fund holdings introduced in 2012.

4 Financial Wealth Estimation Procedure

4.1 Overview of the Procedure

We proceed in five steps. Step 1 estimates mutual fund holdings for 1999–2015, applying two distinct methods before and after the 2012 tax reform. Step 2 recovers ISK capital base values for 2012–2015 from flat-rate tax assessments. Step 3 constructs directly held equity values for 1999–2015 from the Euroclear securities registry with multi-source pricing. Step 4 calculates portfolio-weighted returns. Step 5 integrates all components into the final person-year panel for financial wealth and two person-year datasets for returns.

4.2 Detailed Construction

4.2.1 Step 1: Mutual Fund Holdings

The central challenge in measuring fund wealth is that Swedish administrative data do not directly record individual fund positions. The information available changes structurally with the 2012 tax reform, requiring two separate estimation strategies.

Pre-2012: Inference from dividends. Before 2012, Swedish mutual funds effectively distributed dividends to unit-holders on a universal basis, because doing so allowed funds to avoid taxation at the fund level [2]. The KU31 tax form recorded the total dividend received by each individual from each fund ISIN. We exploit this by dividing the reported dividend by the dividend per share from Morningstar to recover the number of shares held:

$$n_{i,j,t}^{\text{Fund}} = \frac{d_{i,j,t}^{\text{Fin,total}}}{d_{j,t}^{\text{Fin}}}. \tag{3}$$

The fund holding value is then $V_{i,j,t}^{\text{Fund}} = n_{i,j,t}^{\text{Fund}} \cdot \text{NAV}_{j,t}^{\text{Fin}}$, using the year-end NAV converted to SEK.[3]

[2] Foreign funds were not subject to the same taxation rule even if available for sale in Sweden.

[3] When a fund has multiple share classes sharing the same ISIN, preference is given to SEK-denominated classes. Non-SEK dividends and NAVs are converted at year-end exchange rates using

Post-2012: Inversion of the flat-rate tax. From income year 2012, Swedish funds are taxed via a flat-rate standard income equal to $\phi^{\text{Fund}} = 0.4\%$ of the fund's beginning-of-year value. The IoT register reports this standard income, allowing direct recovery of the fund value:

$$W_{i,t-1}^{\text{Fund}} = \frac{\text{SI}_{i,t}^{\text{Fund}}}{\phi^{\text{Fund}}}. \tag{4}$$

Because the tax filing for year t reports the beginning-of-year t value (i.e., end-of-year $t-1$), the observation is assigned to year $t-1$. This yields a single total fund wealth figure per individual-year, without asset-level detail.

The two series are combined into a single panel: the dividend-based method is used for 1999–2011 and the flat-rate inversion for 2012–2015.

4.2.2 Step 2: Investment Savings Accounts (ISK)

The *Investeringssparkonto* (ISK) is a tax-advantaged investment account introduced in 2012. Holdings within an ISK are not taxed on realized gains; instead, a flat-rate standard income is levied annually at a rate σ_t^{ISK} tied to the government borrowing rate plus a markup. The IoT register reports this standard income, and we recover the capital base by inverting the tax formula:

$$W_{i,t}^{\text{ISK}} = \frac{\text{SI}_{i,t}^{\text{ISK}}}{\sigma_t^{\text{ISK}}}. \tag{5}$$

The rates are 1.65% (2012), 1.49% (2013), 2.09% (2014), and 0.90% (2015).[4] The capital base represents an average-balance proxy — the arithmetic average of assets plus deposits measured at the beginning of each quarter of year t — rather than an end-of-year point-in-time stock.

4.2.3 Step 3: Directly Held Equities

Unlike funds, equity positions are directly observed in the Euroclear Sweden securities registry at monthly frequency. We take the latest available monthly snapshot for each year as the year-end proxy and value each position as shares held times the year-end price:

$$V_{i,j,t}^{\text{Stock}} = n_{i,j,t}^{\text{Stock}} \cdot P_{j,t}^{\text{Fin}}. \tag{6}$$

Individual-level stock wealth $W_{i,t}^{\text{Stock}}$ is obtained by summing $V_{i,j,t}^{\text{Stock}}$ across all held ISINs.

The main challenge is pricing. We employ a three-tier hierarchy. First, we use year-end prices from Euroclear's own transaction records. Second, for ISINs without a Euroclear price, we use the last available bid price from Finbas within the calendar year, converting non-SEK quotes at year-end exchange rates. Third, for ISINs still unpriced, we substitute

data from Sveriges Riksbank (2025).

[4] A rate of 1.40% applies in 2016, but the panel ends in 2015.

the price of another share class of the same issuer, preferring the more liquid listing.[5] Securities that remain unpriced after all three tiers receive a value of zero.

Between late 2013 and mid-2014, the Euroclear registry has a known gap affecting small nominee-account holdings. We correct this by carrying forward and backward share counts from adjacent quarters, restricted to positions below 500 shares, and add the imputed holdings back to the affected year-end snapshot.[6]

4.2.4 Step 4: Portfolio Returns

We compute year-over-year fund returns as $r_{j,t}^{\text{Fin}} = \text{NAV}_{j,t}^{\text{Fin}}/\text{NAV}_{j,t-1}^{\text{Fin}} - 1$. For stocks, we compute returns as $r_{j,t}^{\text{Fin}} = P_{j,t}^{\text{Fin}}/P_{j,t-1}^{\text{Fin}} - 1$. In both cases, we construct a balanced individual–asset–year panel, setting holdings to zero in years where the individual does not hold the asset. Portfolio-weighted returns are computed as in equation (2).

The calculation covers 2000–2015, with 1999 serving as the base year for lagged values.

4.2.5 Step 5: Integration

Because the three asset components each cover different subsets of individuals and years, they must be combined into a single panel that assigns a complete set of variables to every individual-year observation. The four component datasets are merged at the individual-year level, with zero values assigned for any component in which an individual does not appear, so that total financial wealth is always the sum of its parts as defined in equation (1).

5 Financial Wealth

The final dataset is a person-year panel keyed by individual identifier and calendar year. It covers 1999–2015 for fund and stock wealth, 2012–2015 for ISK accounts, and 2000–2015 for the portfolio returns. Every individual-year observation contains all variables, with zero-filling where a component is absent. The key output measures are the three wealth levels $W_{i,t}^{\text{Fund}}$, $W_{i,t}^{\text{ISK}}$, and $W_{i,t}^{\text{Stock}}$ defined in equation (1); and the portfolio-weighted returns from equation (2), computed separately for funds and stocks.

References

Euroclear Sweden (2015a). Värdepappersinnehavsregistret (Euroclear Sweden securities holdings register), 1999–2015. (accessed in 2020).

[5] When multiple priced share classes exist for the same issuer, a single common-stock price is selected, with preference for B-shares.

[6] The cap at 500 shares targets the small retail positions that dropped out of the registry during this period.

Euroclear Sweden (2015b). VPstomme (Euroclear Sweden security master / issuer file), 2006–2015. (accessed in 2022).

FinBas (2016). FinBas at Swedish House of Finance Research Data Center, 1983–2016. (accessed in 2016).

Morningstar (2009). Stock and Fund Price Data, 1999–2015. (accessed in 2009).

Statistics Sweden (2016). Income Tax Registry, 1968–2016. (accessed in 2019).

Sveriges Riksbank (2025). Swedish Treasury Bill Returns and Exchange Rates, 1999–2024. (accessed in 2025).

A Glossary

Term	**Definition**
Euroclear Sweden	Central securities depository for Swedish financial instruments, maintaining the official registry of ownership.
Finbas	Swedish equity market database maintained by the Swedish House of Finance, providing daily price data for listed securities.
Investeringssparkonto (ISK)	Tax-advantaged investment savings account introduced in Sweden in 2012. Holdings are taxed via a flat-rate standard income rather than on realized capital gains.
IoT (*Inkomst- och Taxeringsregistret*)	Income and Tax Register maintained by Statistics Sweden, containing annual income and tax data for the Swedish population.
KU31	Tax reporting form (*kontrolluppgift*) filed by fund companies to Skatteverket, reporting dividends paid to individual fund-holders per ISIN.
Morningstar	Commercial provider of fund-level data including net asset values, dividends per share, and fund identifiers.
Schablonintäkt	Flat-rate standard income; a notional income computed as a fixed percentage of asset value, used as the tax base for ISK accounts and (from 2012) mutual fund holdings.

Term	Definition
Skatteverket	The Swedish Tax Agency.
SCB (Statistics Sweden)	*Statistiska centralbyrån*, Sweden's central statistical agency.
VPstomme	Euroclear Sweden's quarterly security master file linking ISINs to issuer information, security types, and classifications.